

Foundation (Jalapeno)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $x(x + 5) + 4(x + 5)$
- (B) $x(x + 4) + 5(x + 4)$
- (C) $x^2 + 9x + 20$
- (D) $x(x + 4) + 5x + 4$
- (E) $x^2 + 4x + 5x + 20$

Q2. Factor the expression: $2x^2 + 10x + 3x + 15$

- (A) $2x(x + 5) + 3(x + 5)$
- (B) $2x(x + 3) + 5(x + 3)$
- (C) $2x^2 + 13x + 15$
- (D) $x(2x + 10) + 3x + 15$
- (E) $x(2x + 3) + 5x + 15$

Q3. Factor the expression: $x^2 + 2x + 3x + 6$

- (A) $x(x + 2) + 3(x + 2)$
- (B) $x(x + 3) + 2(x + 3)$
- (C) $x^2 + 5x + 6$
- (D) $x(x + 3) + 2x + 6$
- (E) $x(x + 2) + 3x + 6$

Q4. Factor the expression: $3x^2 + 12x + 4x + 16$

- (A) $3x(x + 4) + 4(x + 4)$
- (B) $3x(x + 2) + 4(x + 4)$
- (C) $3x^2 + 16x + 16$
- (D) $3x(x + 2) + 4x + 16$
- (E) $x(3x + 12) + 4x + 16$

Q5. Factor the expression: $2x^2 + 6x + 5x + 15$

- (A) $2x(x + 3) + 5(x + 3)$
- (B) $2x(x + 5) + 3(x + 5)$
- (C) $2x^2 + 11x + 15$
- (D) $x(2x + 6) + 5x + 15$
- (E) $x(2x + 5) + 3x + 15$

Q6. Verify the factoring: $x^2 + 4x + 3x + 12 = (x + 3)(x + 4)$

- (A) True
- (B) False
- (C) Maybe
- (D) Always
- (E) Never

Q7. Verify the factoring: $2x^2 + 6x + 4x + 12 = 2(x + 2)(x + 3)$

- (A) True
- (B) False
- (C) Maybe
- (D) Always
- (E) Never

Q8. Verify the factoring: $x^2 + 5x + 2x + 10 = (x + 2)(x + 5)$

- (A) True
- (B) False
- (C) Maybe
- (D) Always
- (E) Never

Open Ended Questions (FRQ)

Q9. Factor the expression: $3x^2 + 12x + 5x + 20$

Q10. Factor the expression: $2x^2 + 8x + 3x + 12$ and verify your factoring by multiplication.

Chef's Tips

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- Tip 1: Ensure students show their work and explain their reasoning.
- Tip 2: Encourage students to use the distributive property to verify their factoring.
- Tip 3: Remind students to check their answers by multiplying the factors.